

Math 63: Real Analysis

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Credit Statement: Math Stack Exchange.

Problem 1. Prove that if the real-valued function f on the interval $[a, b]$ is integrable on $[a, b]$ then so is $|f|$, and $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$.

Solution.

Let $f : [a, b] \rightarrow \mathbb{R}$ be an integrable function. Since f is integrable, given any $\varepsilon > 0$ there exists a $\delta > 0$ such that $|S_1(f) - S_2(f)| < \varepsilon$ when $S_1(f)$ and $S_2(f)$ are riemann sums for f corresponding to partitions of $[a, b]$ of width less than δ . Now, let P be any partition with width less than δ , and pick two different tags t_i, s_i . For each subinterval I_i , define tags:

$$t_i^+ = \begin{cases} t_i & \text{if } f(t_i) \geq f(s_i) \\ s_i & \text{if } f(t_i) < f(s_i) \end{cases} \quad t_i^- = \begin{cases} s_i & \text{if } f(t_i) \geq f(s_i) \\ t_i & \text{if } f(t_i) < f(s_i) \end{cases}$$

Then define:

$$S_1(f) = \sum_i f(t_i^+) \cdot |I_i|, \quad S_2(f) = \sum_i f(t_i^-) \cdot |I_i|$$

And the corresponding riemann sums for $|f|$ are:

$$S_1(|f|) = \sum_i |f(t_i^+)| \cdot |I_i| \quad S_2(|f|) = \sum_i |f(t_i^-)| \cdot |I_i|$$

Then,

$$|S_1(|f|) - S_2(|f|)| \leq \sum_i ||f(t_i^+)| - |f(t_i^-)|| \cdot |I_i| \leq \sum_i |f(t_i^+) - f(t_i^-)| \cdot |I_i| = |S_1(f) - S_2(f)| < \varepsilon$$

That is,

$$|S_1(|f|) - S_2(|f|)| \leq |S_1(f) - S_2(f)| < \varepsilon$$

Hence, the same δ works for $|f|$, and $|f|$ is integrable.

Since both f and $|f|$ are integrable, for any ε there exist a δ (note that the same δ works for both functions as we proved in the previous part) such that $\left| S(f) - \int_a^b f(x) dx \right| < \varepsilon$, and

$\left| S(|f|) - \int_a^b |f(x)| dx \right| < \varepsilon$. Moreover, note that for any given partition and tag choice,

$$|S(f)| = \left| \sum_i f(t_i) |I_i| \right| \leq \sum_i |f(t_i)| \cdot |I_i| = S(|f|)$$

Now note that by the triangle inequality,

$$\left| \int_a^b f(x) dx \right| \leq \left| S(f) - \int_a^b f(x) dx \right| + S(f) < \varepsilon + S(|f|)$$

Also,

$$\left| S(|f|) - \int_a^b f(|x|) dx \right| < \varepsilon \implies S(|f|) < \varepsilon + \int_a^b f(|x|) dx$$

Substituting this into the first inequality gives us,

$$\left| \int_a^b f(x) dx \right| < 2\varepsilon + \int_a^b f(|x|) dx$$

That is,

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$$

Hence proved.

Problem 2. Prove that if f is a continuous real-valued function on the interval $[a, b]$ such that $f(x) \geq 0$ for all $x \in [a, b]$ and $f(x) > 0$ for some $x \in [a, b]$ then $\int_a^b f(x) dx > 0$.

Solution.

Let $c \in [a, b]$ be such that $f(c) > 0$. Given that f is a continuous function, it is continuous at c . Let $\varepsilon_0 = f(c)/2$. Then there exists some δ_0 such that if $|x - c| < \delta_0$ then $|f(x) - f(c)| < \varepsilon_0$. Now consider the interval $I = (c - \delta_0, c + \delta_0) \cap [a, b]$. For any $x \in I$, we have

$$f(x) > f(c) - \varepsilon_0 = f(c) - \frac{f(c)}{2} = \frac{f(c)}{2}$$

Since $c \in [a, b]$, I contains c and therefore has positive length. Then for any $\alpha, \beta \in I$ such that $\alpha < \beta$, we have

$$\int_a^b f(x) dx = \int_a^\alpha f(x) dx + \int_\alpha^\beta f(x) dx + \int_\beta^b f(x) dx$$

Since $f(x) \geq 0$ for all $x \in [a, b]$ we get,

$$\int_a^b f(x) dx \geq \int_\alpha^\beta f(x) dx \geq \frac{f(c)}{2}(\beta - \alpha) > 0$$

Hence proved.

Problem 3. Show that if f is a continuous real-valued function on the interval $[a, b]$ and $f(x) \geq 0$ for all $x \in [a, b]$ then

$$\lim_{n \rightarrow \infty} \left(\int_a^b (f(x))^n dx \right)^{1/n} = \max\{f(x) : x \in [a, b]\}.$$

Solution.

Let f be a continuous real-valued function on the interval $[a, b]$ and $f(x) \geq 0$ for all $x \in [a, b]$. We will prove this by showing that the upper bound equals the lower bound.

Since f is continuous on a closed, bounded interval it attains a maximum $f(c) = M$ at some $c \in [a, b]$. Then, since $f(x) \leq M$ for all $x \in [a, b]$,

$$\int_a^b f(x) dx \leq M(b-a) \implies \int_a^b (f(x))^n dx \leq M^n(b-a)$$

Taking n -th roots gives us,

$$\left(\int_a^b (f(x))^n dx \right)^{1/n} \leq M(b-a)^{1/n}$$

Since f is continuous at c for a given $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$f(x) > M - \varepsilon \quad \forall x \in (c - \delta, c + \delta) \cap [a, b]$$

Let $I_\delta = (c - \delta, c + \delta) \cap [a, b]$, with length ℓ . Then,

$$\int_a^b f(x) dx \geq \int_{I_\delta} f(x) dx \geq (M - \varepsilon) \cdot \ell$$

That is,

$$\int_a^b (f(x))^n dx \geq (M - \varepsilon)^n \cdot \ell$$

Taking n -th roots gives us,

$$\left(\int_a^b (f(x))^n dx \right)^{1/n} \geq (M - \varepsilon) \cdot \ell^{1/n}$$

Combining both bounds gives us,

$$(M - \varepsilon) \cdot \ell \leq \left(\int_a^b (f(x))^n dx \right)^{1/n} \leq M(b-a)^{1/n}$$

Taking limits gives,

$$\lim_{n \rightarrow \infty} (M - \varepsilon) \cdot \ell \leq \lim_{n \rightarrow \infty} \left(\int_a^b (f(x))^n \right)^{1/n} \leq \lim_{n \rightarrow \infty} M(b - a)^{1/n}$$

Note that $n \rightarrow \infty$, $1/n \rightarrow 0$, and $(b - a)^{1/n}$, $\ell^{1/n} \rightarrow 1$. Also, as $\varepsilon \rightarrow 0$, $(M - \varepsilon) \rightarrow M$. So,

$$M \leq \lim_{n \rightarrow \infty} \left(\int_a^b (f(x))^n \right)^{1/n} \leq M$$

Thus,

$$\lim_{n \rightarrow \infty} \left(\int_a^b (f(x))^n \right)^{1/n} = M$$

Hence proved.

Problem 4. Let $[a, b]$ and $[c, d]$ be closed intervals in $\mathbb{R}$ and let f be a continuous real-valued function on $\{(x, y) \in \mathbb{E}^2 : x \in [a, b], y \in [c, d]\}$. Show that the function $g : [c, d] \rightarrow \mathbb{R}$ defined by $g(y) = \int_a^b f(x, y) dx$ for all $y \in [c, d]$ is continuous.

Solution.

We want to show that $g(y) = \int_a^b f(x, y) dx$ is continuous. Let $\varepsilon_g > 0$, y_0 be an arbitrary value in $[c, d]$. Then,

$$|g(y) - g(y_0)| = \left| \int_a^b f(x, y) dx - \int_a^b f(x, y_0) dx \right| = \left| \int_a^b (f(x, y) - f(x, y_0)) dx \right|$$

Note that $f(x, y)$ is continuous on a compact set $[a, b] \times [c, d]$, it is uniformly continuous. Pick $\varepsilon = \varepsilon_g / (b - a) > 0$ then by uniform continuity there exists a $\delta > 0$ such that if $|x - x_0| < \delta$ and $|y - y_0| < \delta$ then $|f(x, y) - f(x_0, y_0)| < \varepsilon$. Since we have $|x - x| = 0 < \delta$ and $|y - y_0| < \delta$,

$$\left| \int_a^b (f(x, y) - f(x, y_0)) dx \right| \leq \int_a^b |f(x, y) - f(x, y_0)| dx < \varepsilon(b - a) = \varepsilon_g$$

Therefore, for all $\varepsilon_g > 0$ there exists a $\delta_g = \delta > 0$ such that if $|y - y_0| < \delta_g$ then $|g(y) - g(y_0)| < \varepsilon_g$. Since ε_g was arbitrary, $g(y)$ is continuous.

Problem 5. A curve in a metric space E is a continuous function f from an interval $[a, b]$ in $\mathbb{R}$ into E ; its length is

$$\text{l.u.b.} \left\{ \sum_{i=1}^N d(f(x_{i-1}), f(x_i)) : x_0, x_1, \dots, x_N \text{ is a partition of } [a, b] \right\},$$

if this l.u.b. exists.

Prove that if $E = \mathbb{R}^n$ and $f(x) = (f_1(x), \dots, f_n(x))$, where $f_1, \dots, f_n : [a, b] \rightarrow \mathbb{R}$ are continuous functions which have continuous derivatives on (a, b) that extend to continuous functions on $[a, b]$, then the curve has length and this length is

$$\int_a^b \sqrt{(f_1'(x))^2 + \dots + (f_n'(x))^2} dx.$$

Solution.

Let S be the set

$$\begin{aligned} S &= \left\{ \sum_{i=1}^N d(f(x_{i-1}), f(x_i)) : x_0, x_1, \dots, x_N \text{ is a partition of } [a, b] \right\} \\ &= \left\{ \sum_{i=1}^N \sqrt{\sum_{j=1}^n (f_j(x_i) - f_j(x_{i-1}))^2} : x_0, x_1, \dots, x_N \text{ is a partition of } [a, b] \right\} \end{aligned}$$

Then, S is not empty, as for the trivial partition $x_0 = a, x_1 = b$,

$$d(f(a), f(b)) = \sqrt{\sum_{j=1}^n (f_j(b) - f_j(a))^2} \in S$$

Also note that since all f_j' are continuous on $[a, b]$, they are bounded. That is for each f_j' there exists some maximum value $M_j = \max_{x \in [a, b]} |f_j'(x)|$, such that for all $x \in [a, b]$, $|f_j'(x)| \leq M_j$.

Then by MVT, for any f_j and any interval $[x_{i-1}, x_i]$ there exists some $c_{ij} \in (x_{i-1}, x_i)$ such that,

$$|f_j(x_i) - f_j(x_{i-1})| = |f_j'(c_{ij})|(x_i - x_{i-1}) \leq M_j(x_i - x_{i-1})$$

Summing over all j gives us

$$d(f(x_{i-1}), f(x_i)) = \sqrt{\sum_{j=1}^n (f_j(x_i) - f_j(x_{i-1}))^2} \leq \sqrt{\sum_{j=1}^n M_j^2} \cdot (x_i - x_{i-1})$$

Summing over all i gives us

$$\sum_{i=1}^N d(f(x_{i-1}), f(x_i)) \leq \sqrt{\sum_{j=1}^n M_j^2} \cdot (b - a)$$

This shows that S is a non-empty set that is bounded from above, hence $\ell = \text{l.u.b. } S$ exists.

Now, let $A = \int_a^b \sqrt{(f'_1(x))^2 + \cdots + (f'_n(x))^2} dx$. We will show $\ell = A$ by proving $\ell \geq A$ and $\ell \leq A$.

First note that for any partition P , adding a point $x_i^* \in [x_{i-1}, x_i]$ cannot decrease the sum, since by the triangle inequality in E^n :

$$d(f(x_{i-1}), f(x_i)) \leq d(f(x_{i-1}), f(x_i^*)) + d(f(x_i^*), f(x_i))$$

Hence for any partition P , any refinement of P has a sum at least as large.

By MVT, for each subinterval $[x_{i-1}, x_i]$ and each j , there exists $c_{ij} \in (x_{i-1}, x_i)$ such that:

$$\sum_{i=1}^N d(f(x_{i-1}), f(x_i)) = \sum_{i=1}^N \sqrt{\sum_{j=1}^n (f'_j(c_{ij}))^2} \cdot (x_i - x_{i-1}) = \sum_{i=1}^N h(c_{ij})(x_i - x_{i-1})$$

where $h(x) = \sqrt{\sum_{j=1}^n (f'_j(x))^2}$. Note that this would be a valid riemann sum for $\int_a^b \sqrt{\sum_{j=1}^n (f'_j(c_{ij}))^2} dx$ except that the c_{ij} depend on j and hence there the c_{ij} are not the same for all j which is required. However, since h is continuous on $[a, b]$ as a composition of continuous functions, hence is uniformly continuous on the compact set $[a, b]$, and we can show that this sum is close to a true riemann sum.

Let $\varepsilon > 0$. Since h is integrable with integral A , there exists $\delta_1 > 0$ such that for any partition of width less than δ_1 and any tag choice t_i :

$$\left| \sum_{i=1}^N h(t_i)(x_i - x_{i-1}) - A \right| < \varepsilon$$

By uniform continuity of h , there exists $\delta_2 > 0$ such that if $|x - y| < \delta_2$ then

$$|h(x) - h(y)| < \varepsilon / (b - a)$$

Then for any partition of width less than δ_2 , any c_{ij} and t_i in the same subinterval satisfy $|c_{ij} - t_i| < \delta_2$, giving:

$$\left| \sum_{i=1}^N h(c_{ij})(x_i - x_{i-1}) - \sum_{i=1}^N h(t_i)(x_i - x_{i-1}) \right| \leq \frac{\varepsilon}{b - a} \sum_{i=1}^N (x_i - x_{i-1}) = \varepsilon$$

Let $\delta = \min(\delta_1, \delta_2)$. Then for any partition of width less than δ , by the triangle inequality:

$$\left| \sum_{i=1}^N d(f(x_{i-1}), f(x_i)) - A \right| \leq \left| \sum_{i=1}^N h(c_{ij})(x_i - x_{i-1}) - \sum_{i=1}^N h(t_i)(x_i - x_{i-1}) \right| + \left| \sum_{i=1}^N h(t_i)(x_i - x_{i-1}) - A \right| < 2\varepsilon$$

That is, for any partition of width less than δ :

$$A - 2\varepsilon < \sum_{i=1}^N d(f(x_{i-1}), f(x_i)) < A + 2\varepsilon$$

Showing $\ell \geq A$:

By definition of ℓ as a least upper bound, there exists a partition P^* with sum within ε of ℓ . Refine P^* to a partition P^{**} of width less than δ . Since refinement does not decrease the sum:

$$\ell \geq \sum_{P^{**}} d(f(x_{i-1}), f(x_i)) \geq \sum_{P^*} d(f(x_{i-1}), f(x_i)) > A - 2\varepsilon$$

Since this holds for any $\varepsilon > 0$, we can rescale $\varepsilon = \varepsilon/2$ we conclude $\ell \geq A$

Showing $\ell \leq A$:

Again by definition of ℓ , there exists a partition P^* with sum within ε of ℓ . Refine P^* to a partition P^{**} of width less than δ . Since refinement does not decrease the sum:

$$\ell - \varepsilon < \sum_{P^{**}} d(f(x_{i-1}), f(x_i)) \leq \sum_{P^*} d(f(x_{i-1}), f(x_i)) < A + 2\varepsilon$$

Hence $\ell < A + 3\varepsilon$. Since this holds for any $\varepsilon > 0$, we can rescale $\varepsilon = \varepsilon/3$ and we conclude $\ell \leq A$. Therefore $\ell \geq A$ and $\ell \leq A$,

$$\ell = A = \int_a^b \sqrt{\sum_{j=1}^n (f'_j(x))^2} dx$$